Football Analytics API Documentation

Version: 1.0.0
Base URL: `http://localhost:8000/api/v1`
Interactive Docs: `http://localhost:8000/docs` (Swagger UI)

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1. Overview

The Football Analytics API provides a comprehensive set of RESTful endpoints for managing and analysing Premier League football data. The API supports full CRUD operations on three core data models (Teams, Players, Matches) and offers five advanced analytics endpoints for statistical analysis.

All responses are returned in **JSON** format. The API follows standard HTTP status codes and includes descriptive error messages for all failure cases.

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2. Authentication

The API uses **JWT (JSON Web Token)** authentication based on the OAuth2 Bearer Token scheme.

Public endpoints (all GET requests) do not require authentication.
Protected endpoints (POST, PUT, DELETE) require a valid JWT token in the `Authorization` header.

2.1 Register a New User

Endpoint: `POST /api/v1/auth/register`

Request Body:

```
{
  "username": "newuser",
  "email": "user@example.com",
  "password": "securepassword"
}
```

Response (201 Created):

```
{
  "id": 3,
  "username": "newuser",
  "email": "user@example.com",
  "is_active": true,
  "is_admin": false,
  "created_at": "2024-01-15T10:30:00"
}
```

Error Responses:

Status Code	Description
409 Conflict	Username or email already registered
422 Unprocessable Entity	Invalid input (e.g., password too short)

2.2 Login

Endpoint: `POST /api/v1/auth/login`

Request Body (form-encoded):

```
username=admin&password=admin123
```

Response (200 OK):

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer"
}
```

Usage: Include the token in subsequent requests:

```
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Error Responses:

Status Code	Description
401 Unauthorized	Incorrect username or password
403 Forbidden	User account is deactivated

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3. Teams Endpoints

3.1 List Teams

Endpoint: `GET /api/v1/teams/`

Query Parameters:

Parameter	Type	Default	Description
`page`	int	1	Page number (min: 1)
`page_size`	int	20	Items per page (1-100)
`search`	string	null	Search by team name (case-insensitive)
`league`	string	null	Filter by league
`city`	string	null	Filter by city

Example Request:

```
GET /api/v1/teams/?page=1&page_size=5&search=Manchester
```

Response (200 OK):

```
{
  "total": 2,
  "page": 1,
  "page_size": 5,
  "teams": [
    {
      "id": 1,
      "name": "Manchester City",
      "short_name": "MCI",
      "founded_year": 1880,
      "stadium": "Etihad Stadium",
      "city": "Manchester",
      "country": "England",
      "league": "Premier League",
      "manager": "Pep Guardiola",
      "budget_millions": 200.0,
      "created_at": "2024-01-01T00:00:00",
      "updated_at": null
    }
  ]
}
```

3.2 Get Team by ID

Endpoint: `GET /api/v1/teams/{team_id}`

Response (200 OK): Single team object as shown above.

Error: 404 Not Found if team does not exist.

3.3 Create Team (Auth Required)

Endpoint: `POST /api/v1/teams/`

Request Body:

```
{
  "name": "Leeds United",
  "short_name": "LEE",
  "city": "Leeds",
  "stadium": "Elland Road",
  "manager": "Daniel Farke",
  "budget_millions": 45.0
}
```

Response (201 Created): The created team object with assigned ID.

Error Responses:

Status Code	Description
401 Unauthorized	No valid authentication token
409 Conflict	Team name already exists
422 Unprocessable Entity	Invalid input data

3.4 Update Team (Auth Required)

Endpoint: `PUT /api/v1/teams/{team_id}`

Supports partial updates - only include fields to be changed.

Request Body:

```
{
  "manager": "New Manager",
  "budget_millions": 120.0
}
```

Response (200 OK): Updated team object.

3.5 Delete Team (Auth Required)

Endpoint: `DELETE /api/v1/teams/{team_id}`

Response: 204 No Content on success.

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4. Players Endpoints

4.1 List Players

Endpoint: `GET /api/v1/players/`

Query Parameters:

Parameter	Type	Default	Description
`page`	int	1	Page number
`page_size`	int	20	Items per page (1-100)
`search`	string	null	Search by player name
`team_id`	int	null	Filter by team
`position`	string	null	Filter by position
`nationality`	string	null	Filter by nationality
`min_goals`	int	null	Minimum goals scored
`sort_by`	string	“name”	Sort field: name, goals, assists, appearances, market_value_millions
`sort_order`	string	“asc”	Sort order: asc or desc

Example Request:

```
GET /api/v1/players/?
sort_by=goals&sort_order=desc&position=Forward&page_size=5
```

Response (200 OK):

```
{
  "total": 15,
  "page": 1,
  "page_size": 5,
  "players": [
    {
      "id": 1,
      "name": "Erling Haaland",
      "age": 23,
      "nationality": "Norway",
      "position": "Forward",
      "jersey_number": 9,
      "goals": 27,
      "assists": 5,
      "appearances": 31,
      "minutes_played": 2650,
      "yellow_cards": 3,
      "red_cards": 0,
      "market_value_millions": 180.0,
      "team_id": 1,
      "created_at": "2024-01-01T00:00:00",
      "updated_at": null
    }
  ]
}
```

4.2 Get Player by ID

Endpoint: `GET /api/v1/players/{player_id}`

4.3 Create Player (Auth Required)

Endpoint: `POST /api/v1/players/`

Request Body:

```
{
  "name": "Jude Bellingham",
  "age": 20,
  "nationality": "England",
  "position": "Midfielder",
  "jersey_number": 5,
  "goals": 12,
  "assists": 8,
  "appearances": 28,
  "team_id": 1
}
```

4.4 Update Player (Auth Required)

Endpoint: `PUT /api/v1/players/{player_id}`

4.5 Delete Player (Auth Required)

Endpoint: `DELETE /api/v1/players/{player_id}`

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5. Matches Endpoints

5.1 List Matches

Endpoint: `GET /api/v1/matches/`

Query Parameters:

Parameter	Type	Default	Description
`page`	int	1	Page number
`page_size`	int	20	Items per page
`season`	string	null	Filter by season (e.g., “2023-2024”)
`team_id`	int	null	Filter by team (home or away)
`date_from`	date	null	Filter from date (YYYY-MM-DD)
`date_to`	date	null	Filter to date (YYYY-MM-DD)
`matchday`	int	null	Filter by matchday

Example Request:

```
GET /api/v1/matches/?season=2023-2024&team_id=1&page_size=3
```

Response (200 OK):

```json
{
  "total": 38,
  "page": 1,
  "page_size": 3,
  "matches": [
    {
      "id": 1,
      "season": "2023-2024",
      "matchday": 1,
      "match_date": "2023-08-14",
      "home_team_id": 1,
      "away_team_id": 2,
      "home_goals": 0,
      "away_goals": 2,
      "home_possession": 55.3,
      "away_possession": 44.7,
      "home_shots": 12,
      "away_shots": 8,
      "home_shots_on_target": 4,
      "away_shots_on_target": 5,
      "home_corners": 7,
      "away_corners": 3,
      "home_fouls": 10,
      "away_fouls": 12,
      "referee": "Michael Oliver",
      "venue": "Etihad Stadium",
      "attendance": 53400,
      "created_at": "2024-01-01T00:00:00",
      "updated_at": null
    }
  ]
}
```


5.2 - 5.5 Get, Create, Update, Delete

Follow the same pattern as Teams and Players endpoints.

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6. Analytics Endpoints

6.1 League Table

Endpoint: `GET /api/v1/analytics/league-table?season=2023-2024`

Generates a complete league standings table with points calculation (3 for win, 1 for draw, 0 for loss), goal difference, and recent form.

Response (200 OK):

```
{
  "season": "2023-2024",
  "table": [
    {
      "position": 1,
      "team_id": 1,
      "team_name": "Manchester City",
      "played": 38,
      "won": 28,
      "drawn": 7,
      "lost": 3,
      "goals_for": 96,
      "goals_against": 34,
      "goal_difference": 62,
      "points": 91,
      "form": "WWDWW"
    }
  ]
}
```

6.2 Team Performance

Endpoint: `GET /api/v1/analytics/team-performance/{team_id}?season=2023-2024`

Returns detailed performance metrics including win rate, average goals, possession stats, clean sheets, and separate home/away records.

Response (200 OK):

```
{
  "team_id": 1,
  "team_name": "Manchester City",
  "season": "2023-2024",
  "total_matches": 38,
  "wins": 28,
  "draws": 7,
  "losses": 3,
  "goals_scored": 96,
  "goals_conceded": 34,
  "goal_difference": 62,
  "points": 91,
  "win_rate": 73.7,
  "avg_goals_scored": 2.53,
  "avg_goals_conceded": 0.89,
  "avg_possession": 63.2,
  "avg_shots": 16.8,
  "clean_sheets": 18,
  "home_record": {"wins": 16, "draws": 2, "losses": 1},
  "away_record": {"wins": 12, "draws": 5, "losses": 2}
}
```

6.3 Player Rankings

Endpoint: `GET /api/v1/analytics/player-rankings?category=goals&limit=10`

Query Parameters:

Parameter	Type	Description
`category`	string	**Required**. One of: `goals`, `assists`, `appearances`, `market_value`
`limit`	int	Number of players (default: 10, max: 50)
`position`	string	Optional position filter
`team_id`	int	Optional team filter

Response (200 OK):

```
{
  "category": "goals",
  "rankings": [
    {
      "rank": 1,
      "player_id": 1,
      "player_name": "Erling Haaland",
      "team_name": "Manchester City",
      "value": 27.0,
      "appearances": 31
    }
  ]
}
```

6.4 Head-to-Head Comparison

Endpoint: `GET /api/v1/analytics/head-to-head?team1_id=1&team2_id=2`

Response (200 OK):

```
{
  "team1_id": 1,
  "team1_name": "Manchester City",
  "team2_id": 2,
  "team2_name": "Arsenal",
  "total_matches": 2,
  "team1_wins": 1,
  "team2_wins": 1,
  "draws": 0,
  "team1_goals": 3,
  "team2_goals": 4,
  "matches": [
    {
      "match_id": 1,
      "date": "2023-08-14",
      "home_team": "Manchester City",
      "away_team": "Arsenal",
      "home_goals": 0,
      "away_goals": 2,
      "result": "Arsenal Win"
    }
  ]
}
```

6.5 Season Summary

Endpoint: `GET /api/v1/analytics/season-summary?season=2023-2024`

Response (200 OK):

```
{
  "season": "2023-2024",
  "total_matches": 380,
  "total_goals": 1115,
  "avg_goals_per_match": 2.93,
  "home_wins": 147,
  "away_wins": 144,
  "draws": 89,
  "home_win_percentage": 38.7,
  "most_goals_match": {
    "match_id": 14,
    "home_team": "Manchester City",
    "away_team": "Everton",
    "score": "3-3",
    "total_goals": 6
  },
  "top_scorer": {
    "player_id": 1,
    "name": "Erling Haaland",
    "goals": 27,
    "team": "Manchester City"
  },
  "top_assister": {
    "player_id": 6,
    "name": "Bukayo Saka",
    "assists": 13,
    "team": "Arsenal"
  }
}
```

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7. Error Handling

All error responses follow a consistent format:

```
{
  "detail": "Descriptive error message"
}
```

Standard HTTP Status Codes

Code	Meaning	When Used
200	OK	Successful GET, PUT requests
201	Created	Successful POST (resource created)
204	No Content	Successful DELETE
400	Bad Request	Invalid input (e.g., same team for home/away)
401	Unauthorized	Missing or invalid authentication token
403	Forbidden	Insufficient permissions
404	Not Found	Resource does not exist
409	Conflict	Duplicate resource (e.g., team name)
422	Unprocessable Entity	Validation error in request body
500	Internal Server Error	Unexpected server error

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8. Data Models

Team

Field	Type	Description
id	integer	Unique identifier (auto-generated)
name	string	Team full name (required, unique)
short_name	string	Short code (e.g., “MCI”)
founded_year	integer	Year founded
stadium	string	Home stadium name
city	string	City location
country	string	Country (default: “England”)
league	string	League name (default: “Premier League”)
manager	string	Current manager
budget_millions	float	Transfer budget in millions

Player

Field	Type	Description
id	integer	Unique identifier
name	string	Player full name (required)
age	integer	Player age (15-50)
nationality	string	Nationality
position	string	Playing position
jersey_number	integer	Shirt number (1-99)
goals	integer	Total goals scored
assists	integer	Total assists
appearances	integer	Total appearances
minutes_played	integer	Total minutes played
yellow_cards	integer	Yellow cards received
red_cards	integer	Red cards received
market_value_millions	float	Market value in millions
team_id	integer	Foreign key to Team

Match

Field	Type	Description
id	integer	Unique identifier
season	string	Season (e.g., “2023-2024”)
matchday	integer	Matchday number
match_date	date	Date of match
home_team_id	integer	Foreign key to home Team
away_team_id	integer	Foreign key to away Team
home_goals	integer	Home team goals
away_goals	integer	Away team goals
home_possession	float	Home possession %
away_possession	float	Away possession %
home_shots	integer	Home total shots
away_shots	integer	Away total shots
home_shots_on_target	integer	Home shots on target
away_shots_on_target	integer	Away shots on target
home_corners	integer	Home corner kicks
away_corners	integer	Away corner kicks
home_fouls	integer	Home fouls
away_fouls	integer	Away fouls
referee	string	Match referee
venue	string	Match venue
attendance	integer	Match attendance